INTERNATIONAL SEARCH REPORT

International application No.
PCT/US21/72513

A. CLASSIFICATION OF SUBJECT MATTER

IPC - G01N 33/574; A61K 39/00; A61P 35/00; C07K 16/30; C07K 16/28; A61K 45/06 (2021.01)

CPC - A61P 35/00; A61K 2039/505; A61K 2039/54; A61K 45/06; C07K 16/30; C07K 16/44; C07K 16/18; A61K 2039/545; A61K 2039/507; A61K 2300/00

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)
See Search History document

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched
See Search History document

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)
See Search History document

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	WO 2018/022933 A1 (OBI PHARMA, INC, ET AL.) 01 February 2018; abstract; paragraphs [005]-[006], [017], [049], [059], [075], [097]	1-25, 33-40
A	WO 2017/185089 A2 (OBI PHARMA, INC., ET AL.) 26 October 2017; entire document	1-25, 33-40
A	WO 2017/004150 A1 (THE JOHNS HOPKINS UNIVERSITY, ET AL.) 05 January 2017; entire document	1-25, 33-40

☐ Further documents are listed in the continuation of Box C. ☐ See patent family annex.

* Special categories of cited documents:
"A" document defining the general state of the art which is not considered to be of particular relevance
"D" document cited by the applicant in the international application
"E" earlier application or patent but published on or after the international filing date
"L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)
"O" document referring to an oral disclosure, use, exhibition or other means
"P" document published prior to the international filing date but later than the priority date claimed
"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention
"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone
"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art
"&" document member of the same patent family

Date of the actual completion of the international search	Date of mailing of the international search report
10 March 2022 (10.03.2022)	24 March 2022 (24.03.2022)
Name and mailing address of the ISA/US Mail Stop PCT, Attn: ISA/US, Commissioner for Patents P.O. Box 1450, Alexandria, Virginia 22313-1450 Facsimile No. 571-273-8300	Authorized officer Shane Thomas Telephone No. PCT Helpdesk: 571-272-4300

INTERNATIONAL SEARCH REPORT

International application No.
PCT/US21/72513

Box No. II Observations where certain claims were found unsearchable (Continuation of item 2 of first sheet)

This international search report has not been established in respect of certain claims under Article 17(2)(a) for the following reasons:

1. ☐ Claims Nos.:
because they relate to subject matter not required to be searched by this Authority, namely:

2. ☐ Claims Nos.:
because they relate to parts of the international application that do not comply with the prescribed requirements to such an extent that no meaningful international search can be carried out, specifically:

3. ☐ Claims Nos.:
because they are dependent claims and are not drafted in accordance with the second and third sentences of Rule 6.4(a).

Box No. III Observations where unity of invention is lacking (Continuation of item 3 of first sheet)

This International Searching Authority found multiple inventions in this international application, as follows:
-***-Please See Supplemental Page-***-

1. ☐ As all required additional search fees were timely paid by the applicant, this international search report covers all searchable claims.

2. ☐ As all searchable claims could be searched without effort justifying additional fees, this Authority did not invite payment of additional fees.

3. ☐ As only some of the required additional search fees were timely paid by the applicant, this international search report covers only those claims for which fees were paid, specifically claims Nos.:

4. ☒ No required additional search fees were timely paid by the applicant. Consequently, this international search report is restricted to the invention first mentioned in the claims; it is covered by claims Nos.:
Group I+, Claims 1-40; gastric cancer (cancer); Globo H (Globo series antigen); cyclophosphamide (anti-proliferative agent)

Remark on Protest

☐ The additional search fees were accompanied by the applicant's protest and, where applicable, the payment of a protest fee.

☐ The additional search fees were accompanied by the applicant's protest but the applicable protest fee was not paid within the time limit specified in the invitation.

☐ No protest accompanied the payment of additional search fees.

INTERNATIONAL SEARCH REPORT

International application No.

PCT/US21/72513

-***-Continued From Box No. III: Observations where unity of invention is lacking-***-

This application contains the following inventions or groups of inventions which are not so linked as to form a single general inventive concept under PCT Rule 13.1. In order for all inventions to be examined, the appropriate additional examination fees must be paid.

Group I+, Claims 1-40; gastric cancer (cancer), Globo H (Globo series antigen), cyclophospharnide (anti-proliferative agent) are directed towards a method for treating advanced cancer in a subject comprising administering to the subject in need thereof a therapeutically effective dose of Globo series antigens vaccine and/or cross-reacting with Globo series antigens vaccine and a method for identifying a suitable patient for cancer therapy.

The methods of Claims 1-17 (each in-part), 18-19, 20 (in-part), 21-25, 33-34, 35 (in-part), 36-38, 39 (in-part) and 40 are believed to encompass the first named invention of Group I+ and are the claims that will be searched without fee to the extent that they encompass gastric cancer (first exemplary cancer), Globo H (first exemplary Globo series antigen), and cyclophosphamide (first exemplary anti-proliferative agent).

Applicant is invited to elect additional cancer(s), Globo series antigen(s), and/or and anti-proliferative agent(s), to be searched. Additional cancer(s), Globo series antigen(s), and/or and anti-proliferative agent(s) will be searched upon the payment of additional fees. Applicants must specify the searchable claims that encompass any additionally elected cancer(s), Globo series antigen(s), and/or and anti-proliferative agent(s). Applicants must further indicate, if applicable, the claims which encompass the first named invention, if different than what was indicated above for this group. Failure to clearly identify how any paid additional invention fees are to be applied to the "+" group(s) will result in only the first claimed invention to be searched/examined. An exemplary election would be lung cancer (cancer).

No technical features are shared between the cancers, Globo series antigens, and/or and anti-proliferative agents of Groups I+ and, accordingly, these groups lack unity a priori.

Additionally, even if Groups I+ were considered to share the technical features including: a method for treating advanced cancer in a subject comprising administering to the subject in need thereof a therapeutically effective dose of Globo series antigens vaccine and/or cross-reacting with Globo series antigens vaccine; a method for inducing/enhancing an immune response in a subject, comprising administering to the subject in an immunogenic agent and/or cross-reacting with the immunogenic agent; a method for treating lung cancer comprising administering to a patient in need thereof a therapeutically effective dose of Globo H-DT CRM197 (diphtheria toxin cross-reacting material 197) glycoconjugate vaccine by a route selected from: (a) Administering vaccine two or more times (e.g., 3, 4, 5, 6, 7, 8, 9, 10, 11, 12 or more times); (b) Adjusting time interval and/or dosing amount regimen between two successive administrations; (c) Adjusting routes of administration and/or altering injection locations of administration; or (d) Any combination of the above thereby each administration increases antibody immune response and/or increases antigen-antibody binding affinity; a method for identifying a suitable patient for cancer therapy in need thereof comprising: (a) Administering an effective dose of Globo series antigen vaccine to the patient; (b) Assessing immune response of the patient; (c) Determining the expression of Globo series antigen in each patient; and (d) Classifying suitability of the patient based on an index of immune response and the expression of Globo series antigens; wherein the index is indicative of the patient having a good prognosis of recovery; these shared technical features are previously disclosed by WO 2017 /185089 A2 (OBI PHARMA, INC.) (hereinafter 'Obi').

OBI discloses a method for treating advanced cancer in a subject (a method for treating advanced cancer in a subject; abstract; claim 1, 67) comprising administering to the subject in need thereof a therapeutically effective dose of Globo series antigens vaccine (comprising administering to the subject in need thereof a therapeutically effective dose of Globo series antigens vaccine; Claim 1) and/or cross-reacting with Globo series antigens vaccine; a method for inducing/enhancing an immune response in a subject, comprising administering to the subject in an immunogenic agent and/or cross-reacting with the immunogenic agent (a method for inducing/enhancing an immune response in a subject, comprising administering to the subject in an immunogenic agent; claims 1, 9, 31); a method for treating lung cancer (a method for treating lung cancer; claims 15, 26) comprising administering to a patient in need thereof a therapeutically effective dose of Globo H-DT CRM197 (diphtheria toxin cross-reacting material 197) glycoconjugate vaccine (comprising administering to a patient in need thereof a therapeutically effective dose of Globo H-DT CRM197 glycoconjugate vaccine by a route; Claims 15-18); by a route selected from: (a) Administering vaccine two or more times (e.g., 3, 4, 5, 6, 7, 8, 9, 10, 11, 12 or more times); (b) Adjusting time interval and/or dosing amount regimen between two successive administrations; (c) Adjusting routes of administration and/or altering injection locations of administration; or (d) Any combination of the above thereby each administration increases antibody immune response and/or increases antigen-antibody binding affinity (by a route selected from: (a) Administering vaccine two or more times (e.g., 3, 4, 5, 6, 7, 8, 9, 10, 11, 12 or more times); (b) Adjusting time interval and/or dosing amount regimen between two successive administrations; (c) Adjusting routes of administration and/or altering injection locations of administration; or (d) Any combination of the above thereby each administration increases antibody immune response and/or increases antigen-antibody binding affinity; claim 51); a method for identifying a suitable patient for cancer therapy in need thereof (a method for identifying a suitable patient for cancer therapy in need thereof; claim 57) comprising: (a) Administering an effective dose of Globo series antigen vaccine to the patient (comprising: (a) Administering an effective dose of Globo series antigen vaccine to the patient; claim 57); (b) Assessing immune response of the patient (Assessing immune response of the patient; claim 57); (c) Determining the expression of Globo series antigen in each patient (Determining the expression of Globo series antigen in each patient; claim 57); and (d) Classifying suitability of the patient based on an index of immune response and the expression of Globo series antigens (Classifying suitability of the patient based on an index of immune response and the expression of Globo series antigens; claim 57); wherein the index is indicative of the patient having a good prognosis of recovery (wherein the index is indicative of the patient having a good prognosis of recovery; claim 57).

Since none of the special technical features of the Groups I+ inventions is found in more than one of the inventions, and since all of the shared technical features are previously disclosed by the OBI reference, unity of invention is lacking.